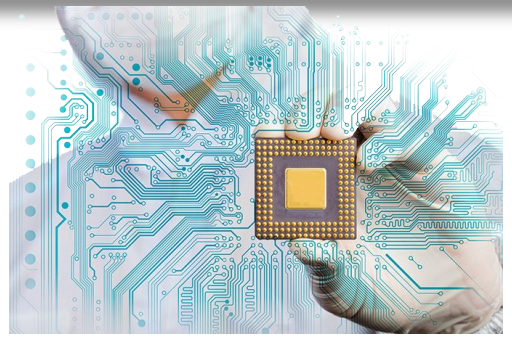
INTRODUCCIÓN A VHDL

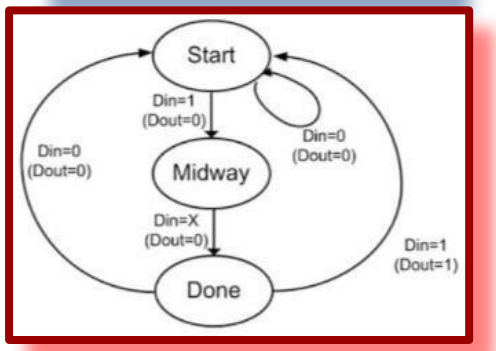
PRINCIPIO DE DISEÑO VLSI
DISEÑO DIGITAL CON TÉCNICAS DE ALTO NIVEL
ÁREA DISPOSITIVOS
DEPTO. DE ELECTRÓNICA Y COMPUTACIÓN
FI.UNMdP



APLICACIONES DE LOS HDLs

¿Para qué sirve un HDL?

MODELAR




```
library ieee;
   use ieee.std_logic_1164.all;

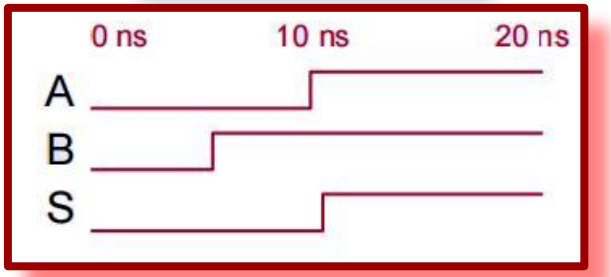
entity nand2 is
   port( a, b : in  std_logic;
         c : out std_logic);
end nand2;

architecture nand2 of nand2 is

begin

   process( a, b )
      variable temp : std_logic;
   begin
      temp := not (a and b);
      if (temp = '1') then
         c <= temp after 6 ns;
      elsif (temp = '0') then
         c <= temp after 5 ns;
      else
         c <= temp after 6 ns;
      end if;
   end process;
end nand2;
```

SIMULAR



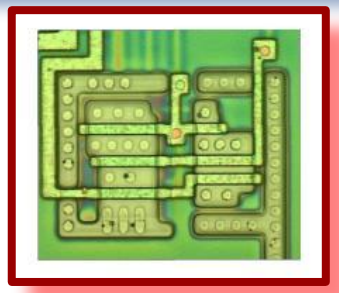

```
-- Clock process
clock: process        -- note: no sens list
begin
   CLK <= '0';
   wait for PER/2;
   inf: loop
      CLK <= not CLK;
      wait for PER/2;
   end loop;
end process;

-- reset process
reset: process        -- note: no sens list
begin
   RSTB <= '0';
   wait for PER*2;
   RSTB <= '1';
   wait;
end process;

-----------------------------------------
-- run process

-- cycle a and c through 4 states and stop
ac: process           -- note: no sens list
begin
   -- initialize values
   A <= '0';
   C <= '0';
```

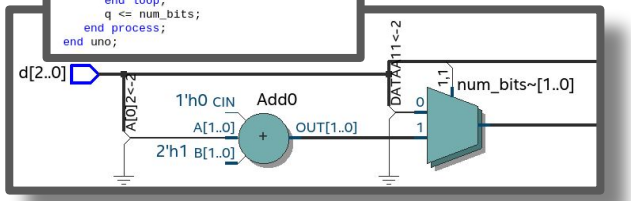
SINTETIZAR

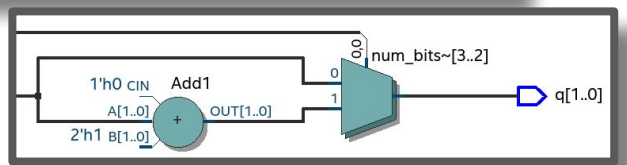


```
library ieee;
   use ieee.std_logic_1164.all;
   use ieee.std_logic_arith.all;

entity ejemplo is
   port ( d : in bit_vector( 2 downto 0 );
          q : out integer range 0 to 3
   );
end ejemplo;

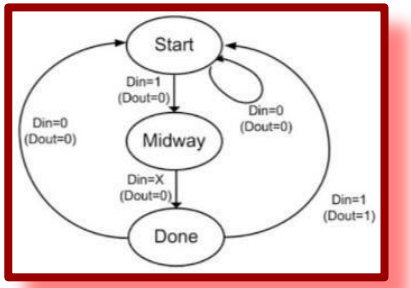
architecture uno of ejemplo is
begin
   process ( d )
      variable num_bits : integer;
   begin
      num_bits := 0;
      for i in d'range loop
         if d( i ) = '1' then
            num_bits := num_bits + 1;
         end if;
      end loop;
      q <= num_bits;
   end process;
end uno;
```





APLICACIONES DE LOS HDLs

MODELAR

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      else
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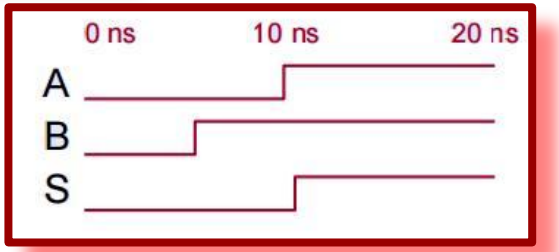
      end if;
   end process;
end nand2;
```

NAND

A B	X
0 0	1
0 1	1
1 0	1
1 1	0

APLICACIONES DE LOS HDLs

SIMULAR

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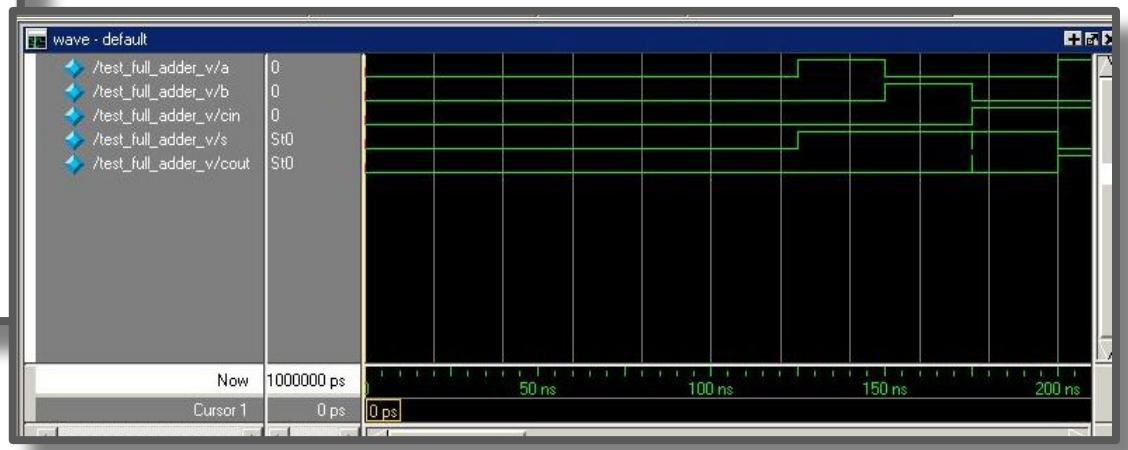
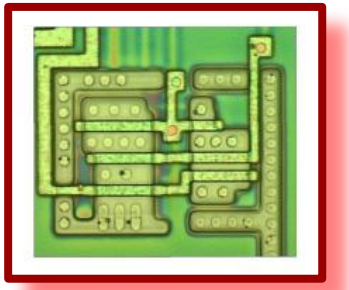
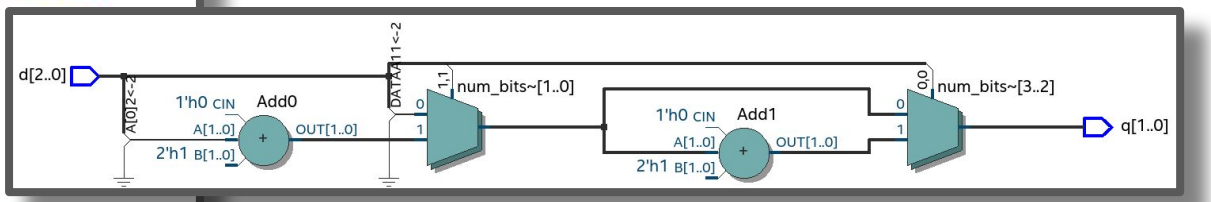


APLICACIONES DE LOS HDLs

SINTETIZAR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         end if;
      end loop;
      q <= num_bits;
   end process;
end uno;
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